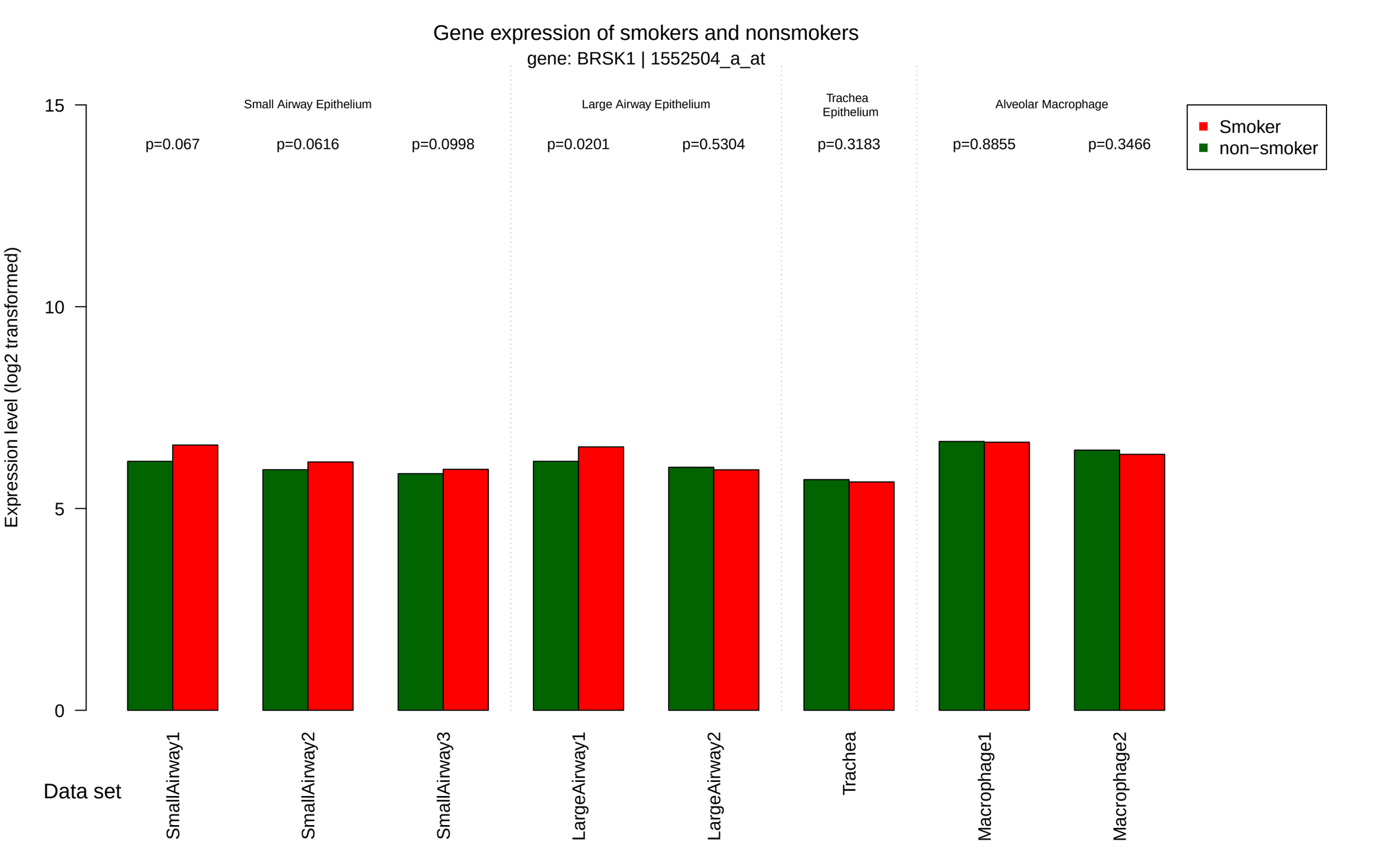
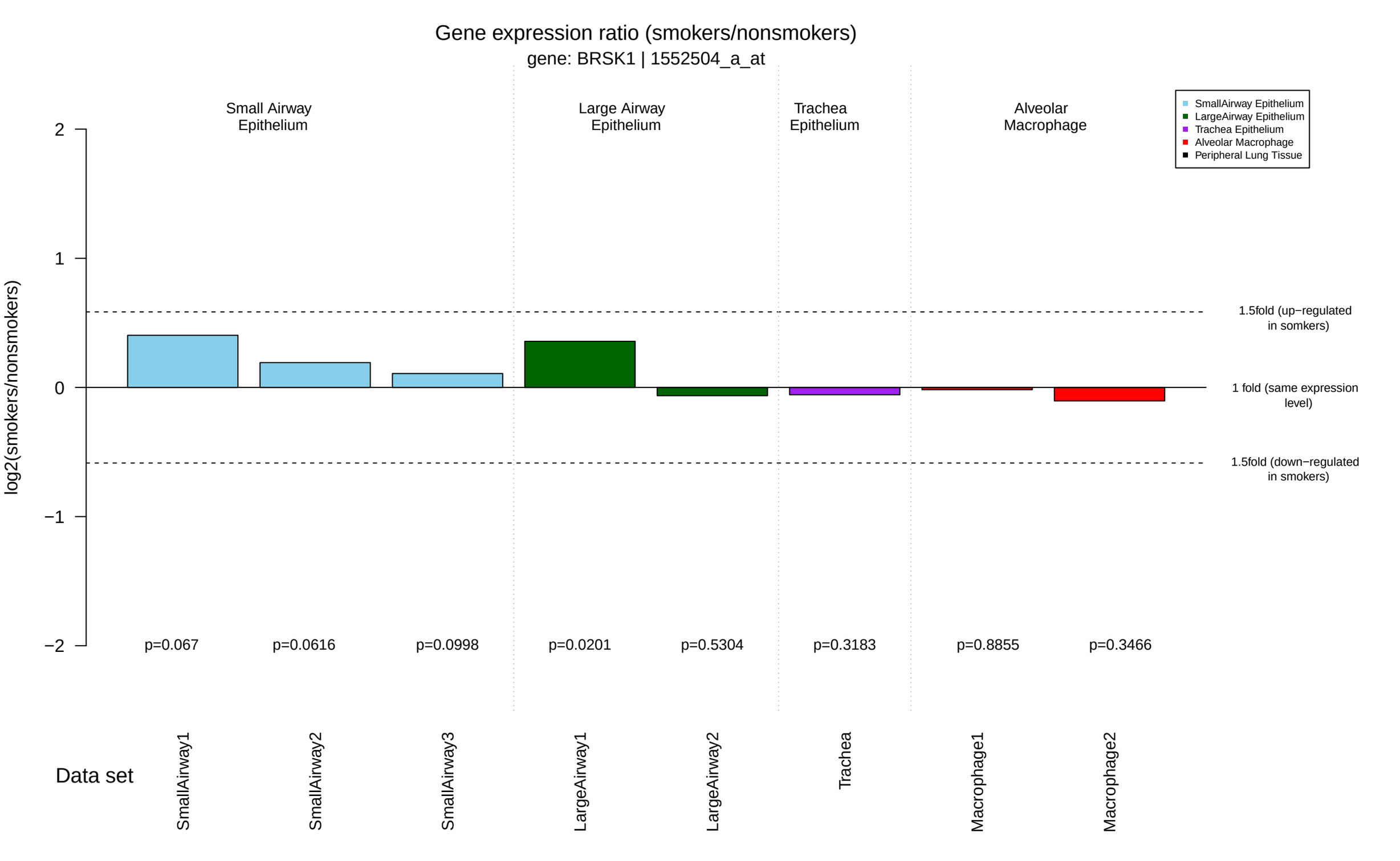
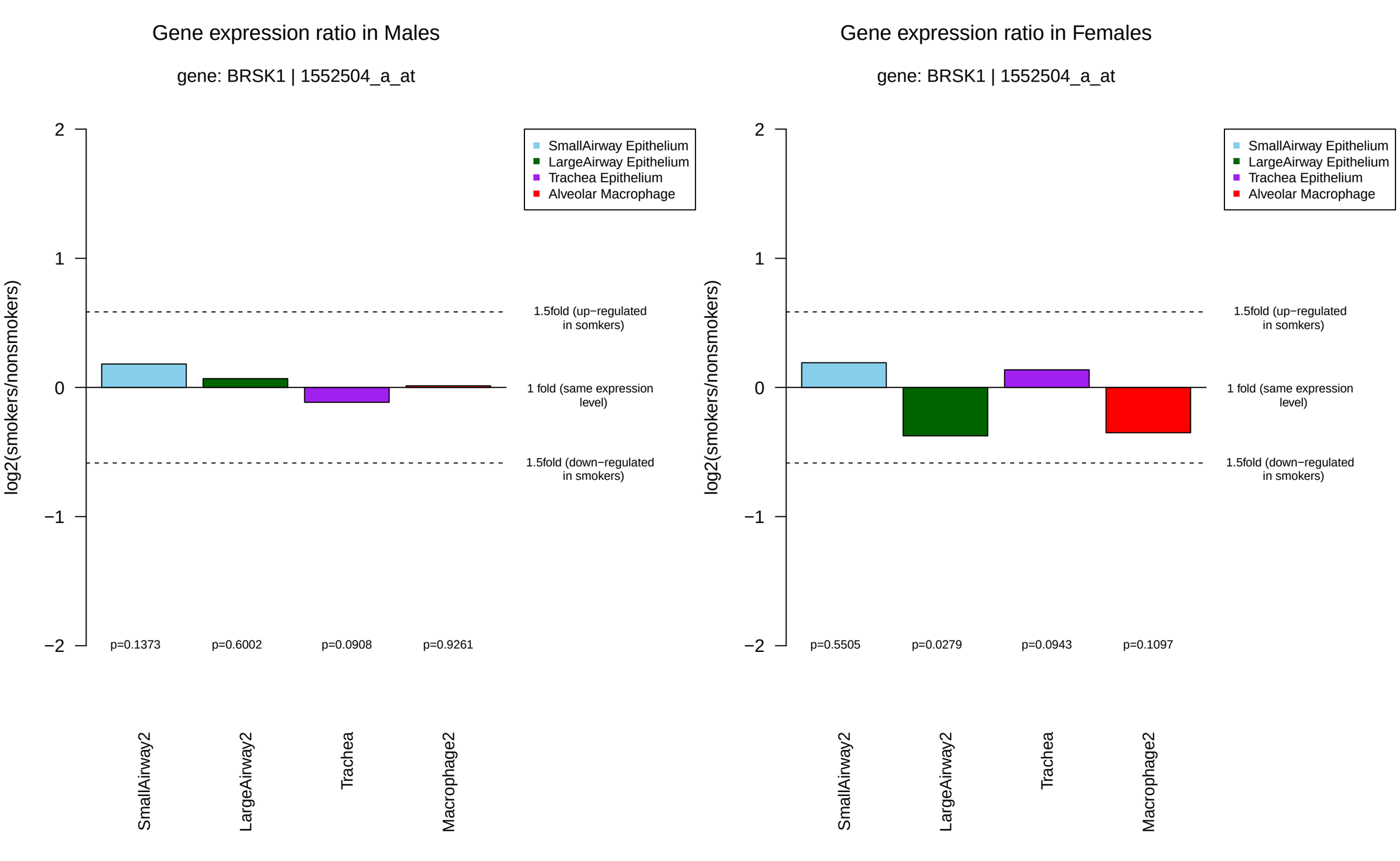
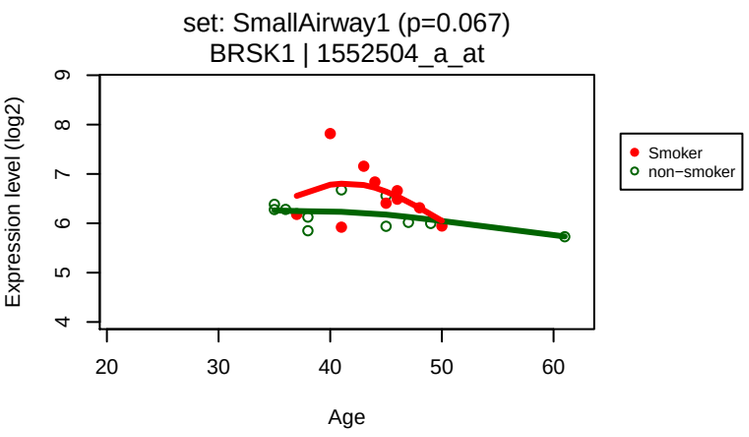
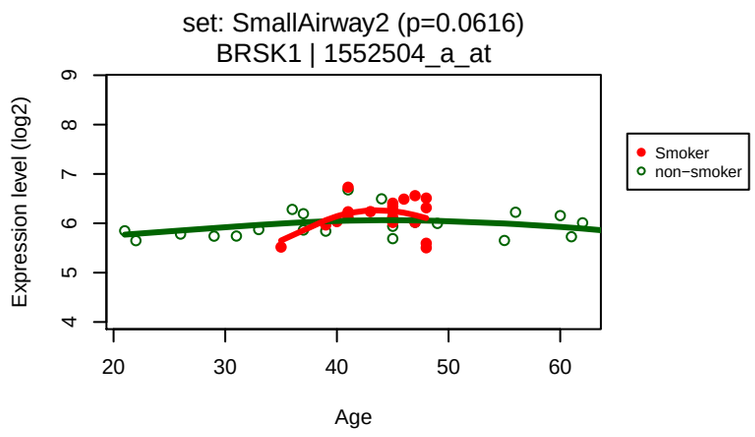
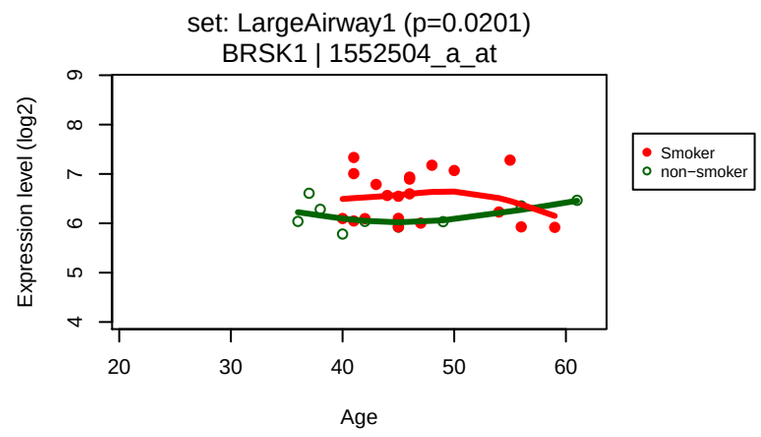

Gene expression of smokers and nonsmokers
gene: BRSK1 | 1552504_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.067
p=0.0616
p=0.0998
p=0.0201
p=0.5304
p=0.3183
p=0.8855
p=0.3466
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

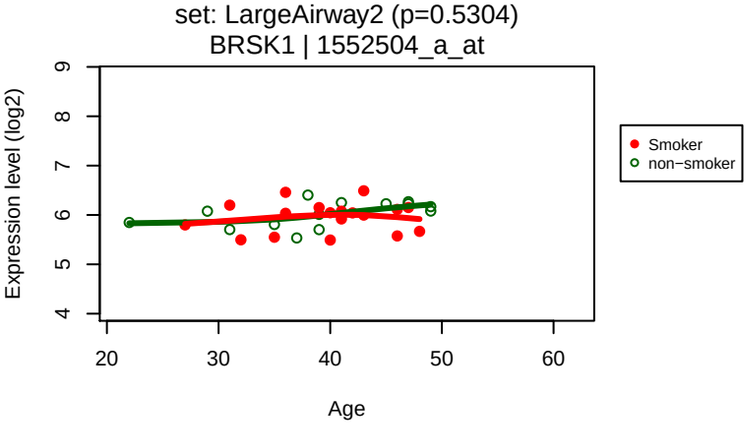

Gene expression ratio (smokers/nonsmokers)
gene: BRSK1 | 1552504_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.067
p=0.0616
p=0.0998
p=0.0201
p=0.5304
p=0.3183
p=0.8855
p=0.3466
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

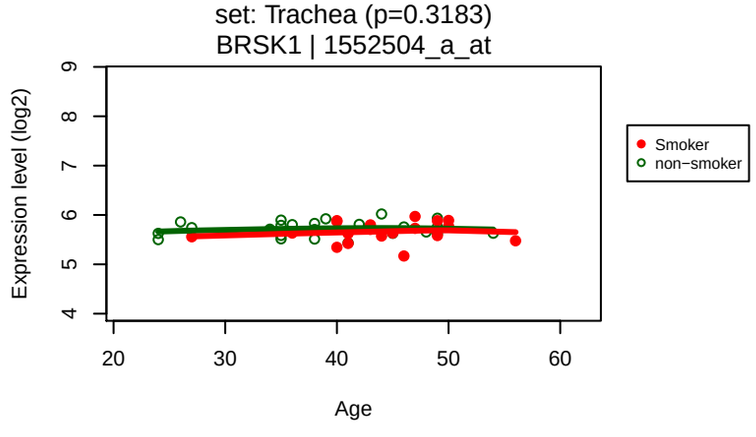

Gene expression ratio in Males
gene: BRSK1 | 1552504_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1373
p=0.6002
p=0.0908
p=0.9261
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRSK1 | 1552504_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5505
p=0.0279
p=0.0943
p=0.1097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

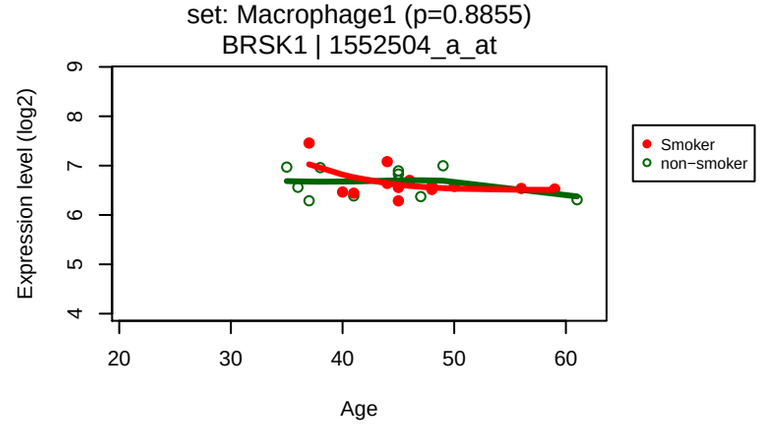

set: SmallAirway1 (p=0.067)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker

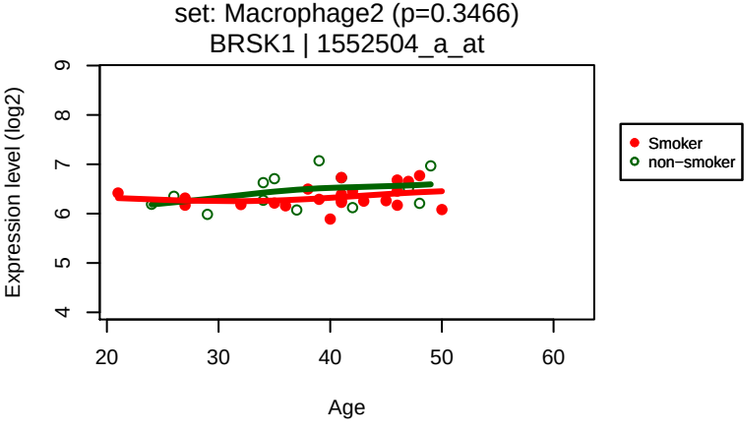

set: SmallAirway2 (p=0.0616)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker

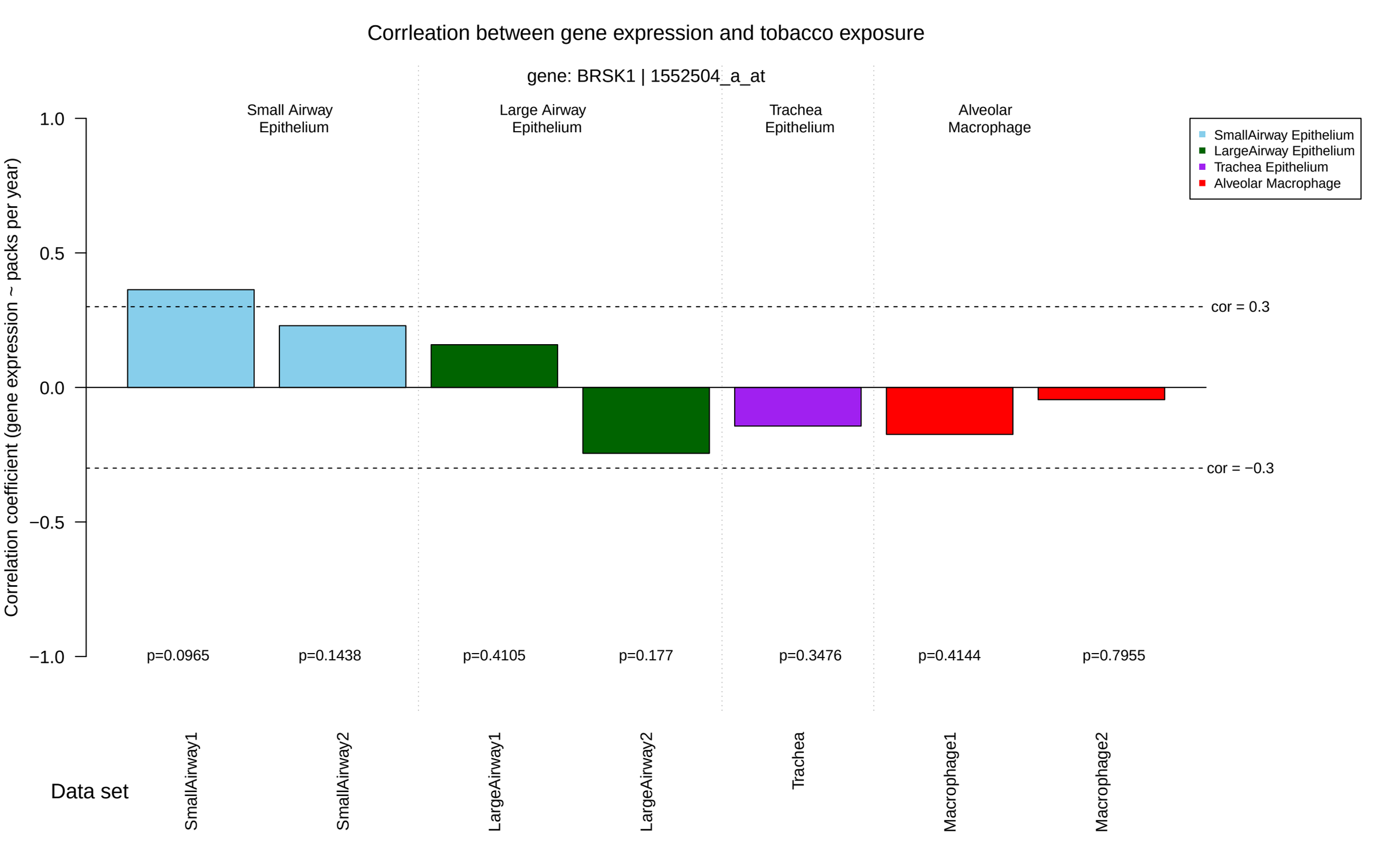

set: LargeAirway1 (p=0.0201)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5304)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3183)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8855)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3466)
BRSK1 | 1552504_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRSK1 | 1552504_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0965
p=0.1438
p=0.4105
p=0.177
p=0.3476
p=0.4144
p=0.7955
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2